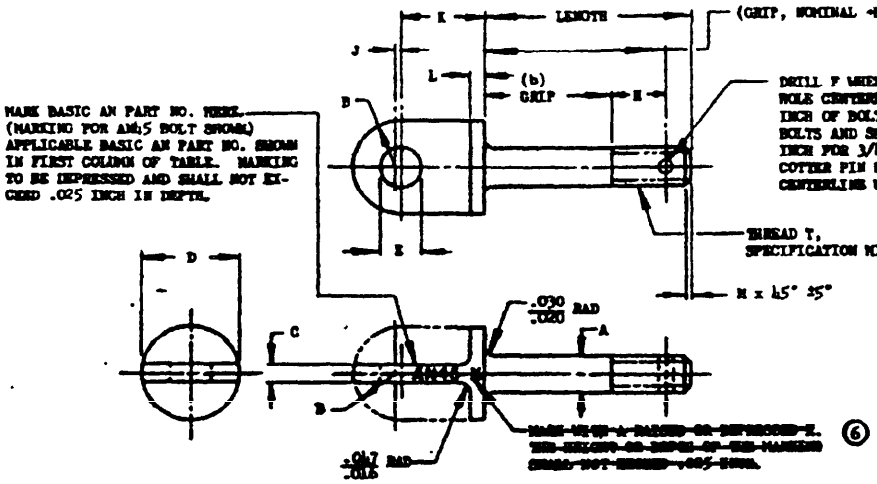
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NOTE: This drawing was approved by joint action of the Air Force and Navy Departments as the Air Force-Navy standard for this product. This drawing supersedes all associated standard drawings for the same product and shall become effective for the procurement of associated supplies, or for use in new designs, not later than 6 months after the latest date of approval shown.

AVAILABLE FROM OSD/ISA

FSC 5306



BASIC AN PART NO.	THREAD T	A DIA MAX	A DIA MIN	B RAD	C ±.005	D DIA	E DIA MAX	E DIA MIN	F ±.010	(a) H REF	J	K	L	M	RATED STRENGTH, POUNDS: ULTIMATE STRENGTH OF BOLT	RATED STRENGTH, POUNDS: YIELD STRENGTH OF BOLT	(a) PIN DIA REF
(c) AN42	NO. 10-32 UNF-3A	.189	.186	7/32	.078	7/16	.194	.190	.078	2/64	1/32	13/32	1/32	1/32	1 250	880	3/16
AN42B							.192	.190									
(c) AN43	1/4-28 UNF-3A	.249	.246	1/4	.125	1/2	.256	.250	.086	5/64	1/32	15/32	5/64	1/32	2 450	1 880	3/16
AN43B							.192	.190									
AN44	5/16-24 UNF-3A	.312	.309	5/16	.188	5/8	.253	.250	.086	2/64	3/64	17/32	3/32	3/64	3 500	3 000	1/4
AN45	5/16-24 UNF-3A	.312	.309	11/32		11/16	.316	.313	.086	2/64		19/32			5 250	4 055	5/16
AN46	3/8-24 UNF-3A	.374	.371	3/8		3/4	.378	.375	.106	7/64		19/32			7 085	5 375	3/8
AN47	7/16-20 UNF-3A	.437	.433	7/16	.250	7/8	.378	.375	.106	11/64	1/16	23/32		3/64	9 200	7 055	3/8
AN48	1/2-20 UNF-3A	.499	.495	1/2	.313	1	.441	.438	.106	13/64	1/16	13/16	1/8	3/64	14 375	11 020	7/16
AN49	9/16-18 UNF-3A	.562	.558	19/32	.375	1-3/16	.503	.500	.141	21/64	5/64	15/16		1/16	20 125	15 430	1/2

(a) REFERENCE DIMENSIONS ARE FOR DESIGN PURPOSES ONLY AND ARE NOT AN INSPECTION REQUIREMENT.

(b) GRIP LENGTH OF BOLTS SHALL BE MEASURED FROM THE UNDERSIDE OF THE HEAD TO THE END OF THE FULL CYLINDRICAL PORTION OF THE SHANK. COMPLETE THREADS SHALL BEGIN WITHIN TWO THREAD PITCH MAXIMUM. TWO THREAD PITCH MAXIMUM MAY CONSIST OF INCOMPLETE THREAD OR EXTRUSION ANGLE.

(c) BOLTS WITH BASIC PART NUMBERS AN42 AND AN43 INACTIVE FOR DESIGN AFTER 4 APRIL 1955. IN LIEU THEREOF, USE BOLTS WITH BASIC PART NUMBERS AN42B AND AN43B.

FOR DEFINITION AND APPLICATION OF DRAWING STATUS NOTES, SEE AMA BULLETIN NO. 337.

MATERIAL: NON-CORROSION-RESISTANT STEEL OR CORROSION-RESISTANT STEEL. SEE PROCUREMENT SPECIFICATION.

⑤ FINISH: SEE PROCUREMENT SPECIFICATION.

ADD A AFTER DASH NUMBER FOR BOLT WITHOUT DRILLED HOLE IN SHANK.

ADD C BEFORE DASH NUMBER FOR CORROSION-RESISTANT STEEL BOLT.

EXAMPLES OF PART NUMBERS: AN46-10 - 3/8-24 NON-CORROSION-RESISTANT STEEL EYE BOLT, 1-5/64 LONG HAVING 7/16 GRIP AND WITH DRILLED HOLE IN SHANK.
AN46-10A - 3/8-24 NON-CORROSION-RESISTANT STEEL EYE BOLT, 1-5/64 LONG HAVING 7/16 GRIP AND WITHOUT DRILLED HOLE IN SHANK.
AN46-C10 - 3/8-24 CORROSION-RESISTANT STEEL EYE BOLT, 1-5/64 LONG HAVING 7/16 GRIP AND WITH DRILLED HOLE IN SHANK.
AN46-C10A - 3/8-24 CORROSION-RESISTANT STEEL EYE BOLT, 1-5/64 LONG HAVING 7/16 GRIP AND WITHOUT DRILLED HOLE IN SHANK.
AN43B-C10A - 1/4-28 CORROSION-RESISTANT STEEL EYE BOLT, 1/32 LONG HAVING 5/16 GRIP AND WITHOUT DRILLED HOLE IN SHANK.

BOLTS SHALL BE FREE FROM ALL MACHINE BURRS AND SLIVERS WHICH MIGHT BECOME DISLODGED UNDER SHANK.

COUNTERSINKING OF DRILLED HOLE IN SHANK IS OPTIONAL. PIN HOLE E SHALL NOT BE COUNTERSUNK.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: FRACTIONS ±1/64.

⑤ FOR CHANGES SEE SHEET 2.

APPROVED 15 Jan 43 REVISED (3) 14 Apr 55 (4) 30 Jun 66 (5) 23 Feb 67 (6) 4 Aug 82

PROCUREMENT SPECIFICATION	AIR FORCE-NAVY AERONAUTICAL STANDARD	AN42 THRU AN49
MIL-B-6812	BOLT, EYE	SHEET 1 OF 2

SUPERSEDES FORMER AIR FORCE AND NAVY STANDARD ISSUES OF AN42 THROUGH AN49.

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DASH NO.	AN42, AN42B GRIP	AN42, AN42B LENGTH +1/32 -1/64	AN43, AN43B GRIP	AN43, AN43B LENGTH +1/32 -1/64	AN44 GRIP	AN44 LENGTH +1/32 -1/64	AN45 GRIP	AN45 LENGTH +1/32 -1/64	AN46 GRIP	AN46 LENGTH +1/32 -1/64	AN47 GRIP	AN47 LENGTH +1/32 -1/64	AN48 GRIP	AN48 LENGTH +1/32 -1/64	AN49 GRIP	AN49 LENGTH +1/32 -1/64
3	1/16	15/32														
4	1/8	17/32	1/16	17/32	1/16	19/32	1/16	19/32								
5	1/4	21/32	3/16	21/32	3/16	23/32	3/16	23/32	1/16	45/64	1/16	23/32				
6	3/8	25/32	5/16	25/32	5/16	27/32	5/16	27/32	3/16	53/64	3/16	27/32	1/16	27/32	1/16	31/32
7	1/2	29/32	7/16	29/32	7/16	31/32	7/16	31/32	5/16	61/64	5/16	31/32	3/16	31/32	1/3	1- 1/32
10	5/8	1- 1/32	9/16	1- 1/32	9/16	1- 3/32	9/16	1- 3/32	7/16	1- 5/64	7/16	1- 3/32	5/16	1- 3/32	1/4	1- 5/32
11	3/4	1- 5/32	11/16	1- 5/32	11/16	1- 7/32	11/16	1- 7/32	9/16	1-13/64	9/16	1- 7/32	7/16	1- 7/32	3/8	1- 9/32
12	7/8	1- 9/32	13/16	1- 9/32	13/16	1-11/32	13/16	1-11/32	11/16	1-21/64	11/16	1-11/32	9/16	1-11/32	1/2	1-13/32
13	1	1-13/32	15/16	1-13/32	15/16	1-15/32	15/16	1-15/32	13/16	1-29/64	13/16	1-15/32	11/16	1-15/32	5/8	1-17/32
14	1-1/8	1-17/32	1- 1/16	1-17/32	1- 1/16	1-19/32	1- 1/16	1-19/32	15/16	1-37/64	15/16	1-19/32	13/16	1-19/32	3/4	1-21/32
15	1-1/4	1-21/32	1- 3/16	1-21/32	1- 3/16	1-23/32	1- 3/16	1-23/32	1- 1/16	1-45/64	1- 1/16	1-23/32	15/16	1-23/32	7/8	1-25/32
16	1-3/8	1-25/32	1- 5/16	1-25/32	1- 5/16	1-27/32	1- 5/16	1-27/32	1- 3/16	1-53/64	1- 3/16	1-27/32	1- 1/16	1-27/32	1	1-29/32
17	1-1/2	1-29/32	1- 7/16	1-29/32	1- 7/16	1-31/32	1- 7/16	1-31/32	1- 5/16	1-61/64	1- 5/16	1-31/32	1- 3/16	1-31/32	1-1/8	2- 1/32
20	1-5/8	2- 1/32	1- 9/16	2- 1/32	1- 9/16	2- 3/32	1- 9/16	2- 3/32	1- 7/16	2- 5/64	1- 7/16	2- 3/32	1- 5/16	2- 3/32	1-1/4	2- 5/32
21	1-3/4	2- 5/32	1-11/16	2- 5/32	1-11/16	2- 7/32	1-11/16	2- 7/32	1- 9/16	2-13/64	1- 9/16	2- 7/32	1- 7/16	2- 7/32	1-3/8	2- 9/32
22	1-7/8	2- 9/32	1-13/16	2- 9/32	1-13/16	2-11/32	1-13/16	2-11/32	1-11/16	2-21/64	1-11/16	2-11/32	1- 9/16	2-11/32	1-1/2	2-13/32
23	2	2-13/32	1-15/16	2-13/32	1-15/16	2-15/32	1-15/16	2-15/32	1-13/16	2-29/64	1-13/16	2-15/32	1-11/16	2-15/32	1-5/8	2-17/32
24	2-1/8	2-17/32	2- 1/16	2-17/32	2- 1/16	2-19/32	2- 1/16	2-19/32	1-15/16	2-37/64	1-15/16	2-19/32	1-13/16	2-19/32	1-3/4	2-21/32
25	2-1/4	2-21/32	2- 3/16	2-21/32	2- 3/16	2-23/32	2- 3/16	2-23/32	2- 1/16	2-45/64	2- 1/16	2-23/32	1-15/16	2-23/32	1-7/8	2-25/32
26	2-3/8	2-25/32	2- 5/16	2-25/32	2- 5/16	2-27/32	2- 5/16	2-27/32	2- 3/16	2-53/64	2- 3/16	2-27/32	2- 1/16	2-27/32	2	2-29/32
27	2-1/2	2-29/32	2- 7/16	2-29/32	2- 7/16	2-31/32	2- 7/16	2-31/32	2- 5/16	2-61/64	2- 5/16	2-31/32	2- 3/16	2-31/32	2-1/8	3- 1/32
30	2-5/8	3- 1/32	2- 9/16	3- 1/32	2- 9/16	3- 3/32	2- 9/16	3- 3/32	2- 7/16	3- 5/64	2- 7/16	3- 3/32	2- 5/16	3- 3/32	2-1/4	3- 5/32
31	2-3/4	3- 5/32	2-11/16	3- 5/32	2-11/16	3- 7/32	2-11/16	3- 7/32	2- 9/16	3-13/64	2- 9/16	3- 7/32	2- 7/16	3- 7/32	2-3/8	3- 9/32
32	2-7/8	3- 9/32	2-13/16	3- 9/32	2-13/16	3-11/32	2-13/16	3-11/32	2-11/16	3-21/64	2-11/16	3-11/32	2- 9/16	3-11/32	2-1/2	3-13/32
33	3	3-13/32	2-15/16	3-13/32	2-15/16	3-15/32	2-15/16	3-15/32	2-13/16	3-29/64	2-13/16	3-15/32	2-11/16	3-15/32	2-5/8	3-17/32
34	3-1/8	3-17/32	3- 1/16	3-17/32	3- 1/16	3-19/32	3- 1/16	3-19/32	2-15/16	3-37/64	2-15/16	3-19/32	2-13/16	3-19/32	2-3/4	3-21/32
35	3-1/4	3-21/32	3- 3/16	3-21/32	3- 3/16	3-23/32	3- 3/16	3-23/32	3- 1/16	3-45/64	3- 1/16	3-23/32	2-15/16	3-23/32	2-7/8	3-25/32
36	3-3/8	3-25/32	3- 5/16	3-25/32	3- 5/16	3-27/32	3- 5/16	3-27/32	3- 3/16	3-53/64	3- 3/16	3-27/32	3- 1/16	3-27/32	3	3-29/32
37	3-1/2	3-29/32	3- 7/16	3-29/32	3- 7/16	3-31/32	3- 7/16	3-31/32	3- 5/16	3-61/64	3- 5/16	3-31/32	3- 3/16	3-31/32	3-1/8	4- 1/32
40	3-5/8	4- 1/32	3- 9/16	4- 1/32	3- 9/16	4- 3/32	3- 9/16	4- 3/32	3- 7/16	4- 5/64	3- 7/16	4- 3/32	3- 5/16	4- 3/32	3-1/8	4- 5/32
41	3-3/4	4- 5/32	3-11/16	4- 5/32	3-11/16	4- 7/32	3-11/16	4- 7/32	3- 9/16	4-13/64	3- 9/16	4- 7/32	3- 7/16	4- 7/32	3-3/8	4- 9/32
42	3-7/8	4- 9/32	3-13/16	4- 9/32	3-13/16	4-11/32	3-13/16	4-11/32	3-11/16	4-21/64	3-11/16	4-11/32	3- 9/16	4-11/32	3-1/2	4-13/32
43	4	4-13/32	3-15/16	4-13/32	3-15/16	4-15/32	3-15/16	4-15/32	3-13/16	4-29/64	3-13/16	4-15/32	3-11/16	4-15/32	3-5/8	4-17/32
44	4-1/8	4-17/32	4- 1/16	4-17/32	4- 1/16	4-19/32	4- 1/16	4-19/32	3-15/16	4-37/64	3-15/16	4-19/32	3-13/16	4-19/32	3-3/4	4-21/32
45	4-1/4	4-21/32	4- 3/16	4-21/32	4- 3/16	4-23/32	4- 3/16	4-23/32	4- 1/16	4-45/64	4- 1/16	4-23/32	3-15/16	4-23/32	3-7/8	4-25/32
46	4-3/8	4-25/32	4- 5/16	4-25/32	4- 5/16	4-27/32	4- 5/16	4-27/32	4- 3/16	4-53/64	4- 3/16	4-27/32	4- 1/16	4-27/32	4	4-29/32
47	4-1/2	4-29/32	4- 7/16	4-29/32	4- 7/16	4-31/32	4- 7/16	4-31/32	4- 5/16	4-61/64	4- 5/16	4-31/32	4- 3/16	4-31/32	4-1/8	5- 1/32
50	4-5/8	5- 1/32	4- 9/16	5- 1/32	4- 9/16	5- 3/32	4- 9/16	5- 3/32	4- 7/16	5- 5/64	4- 7/16	5- 3/32	4- 5/16	5- 3/32	4-1/4	5- 5/32
51	4-3/4	5- 5/32	4-11/16	5- 5/32	4-11/16	5- 7/32	4-11/16	5- 7/32	4- 9/16	5-13/64	4- 9/16	5- 7/32	4- 7/16	5- 7/32	4-3/8	5- 9/32
52	4-7/8	5- 9/32	4-13/16	5- 9/32	4-13/16	5-11/32	4-13/16	5-11/32	4-11/16	5-21/64	4-11/16	5-11/32	4- 9/16	5-11/32	4-1/2	5-13/32
53	5	5-13/32	4-15/16	5-13/32	4-15/16	5-15/32	4-15/16	5-15/32	4-13/16	5-29/64	4-13/16	5-15/32	4-11/16	5-15/32	4-5/8	5-17/32
54	5-1/8	5-17/32	5- 1/16	5-17/32	5- 1/16	5-19/32	5- 1/16	5-19/32	4-15/16	5-37/64	4-15/16	5-19/32	4-13/16	5-19/32	4-3/4	5-21/32
55	5-1/4	5-21/32	5- 3/16	5-21/32	5- 3/16	5-23/32	5- 3/16	5-23/32	5- 1/16	5-45/64	5- 1/16	5-23/32	4-15/16	5-23/32	4-7/8	5-25/32
56	5-3/8	5-25/32	5- 5/16	5-25/32	5- 5/16	5-27/32	5- 5/16	5-27/32	5- 3/16	5-53/64	5- 3/16	5-27/32	5- 1/16	5-27/32	5	5-29/32
57	5-1/2	5-29/32	5- 7/16	5-29/32	5- 7/16	5-31/32	5- 7/16	5-31/32	5- 5/16	5-61/64	5- 5/16	5-31/32	5- 3/16	5-31/32	5-1/8	6- 1/32
60	5-5/8	6- 1/32	5- 9/16	6- 1/32	5- 9/16	6- 3/32	5- 9/16	6- 3/32	5- 7/16	6- 5/64	5- 7/16	6- 3/32	5- 5/16	6- 3/32	5-1/4	6- 5/32
61	5-3/4	6- 5/32	5-11/16	6- 5/32	5-11/16	6- 7/32	5-11/16	6- 7/32	5- 9/16	6-13/64	5- 9/16	6- 7/32	5- 7/16	6- 7/32	5-3/8	6- 9/32
62	5-7/8	6- 9/32	5-13/16	6- 9/32	5-13/16	6-11/32	5-13/16	6-11/32	5-11/16	6-21/64	5-11/16	6-11/32	5- 9/16	6-11/32	5-1/2	6-13/32
63	6	6-13/32	5-15/16	6-13/32	5-15/16	6-15/32	5-15/16	6-15/32	5-13/16	6-29/64	5-13/16	6-15/32	5-11/16	6-15/32	5-5/8	6-17/32
64	6-1/8	6-17/32	6- 1/16	6-17/32	6- 1/16	6-19/32	6- 1/16	6-19/32	5-15/16	6-37/64	5-15/16	6-19/32	5-13/16	6-19/32	5-3/4	6-21/32
65	6-1/4	6-21/32	6- 3/16	6-21/32	6- 3/16	6-23/32	6- 3/16	6-23/32	6- 1/16	6-45/64	6- 1/16	6-23/32	5-15/16	6-23/32	5-7/8	6-25/32
66	6-3/8	6-25/32	6- 5/16	6-25/32	6- 5/16	6-27/32	6- 5/16	6-27/32	6- 3/16	6-53/64	6- 3/16	6-27/32	6- 1/16	6-27/32	6	6-29/32
67	6-1/2	6-29/32	6- 7/16	6-29/32	6- 7/16	6-31/32	6- 7/16	6-31/32	6- 5/16	6-61/64	6- 5/16	6-31/32	6- 3/16	6-31/32	6-1/8	7- 1/32
70	6-5/8	7- 1/32	6- 9/16	7- 1/32	6- 9/16	7- 3/32	6- 9/16	7- 3/32	6- 7/16	7- 5/64	6- 7/16	7- 3/32	6- 5/16	7- 3/32	6-1/4	7- 5/32
71	6-3/4	7- 5/32	6-11/16	7- 5/32	6-11/16	7- 7/32	6-11/16	7- 7/32	6- 9/16	7-13/64	6- 9/16	7- 7/32	6- 7/16	7- 7/32	6-3/8	7- 9/32
72	6-7/8	7- 9/32	6-13/16	7- 9/32	6-13/16	7-11/32	6-13/16	7-11/32	6-11/16	7-21/64	6-11/16	7-11/32	6- 9/16	7-11/32	6-1/2	7-13/32
73	7	7-13/32	6-15/16	7-13/32	6-15/16	7-15/32	6-15/16	7-15/32	6-13/16	7-29/64	6-13/16	7-15/32	6-11/16	7-15/32	6-5/8	7-17/32
74	7-1/8	7-17/32	7- 1/16	7-17/32	7- 1/16	7-19/32	7- 1/16	7-19/32	6-15/16	7-37/64	6-15/16	7-19/32	6-13/16	7-19/32	6-3/4	7-21/32
75	7-1/4	7-21/32	7- 3/16	7-21/32	7- 3/16	7-23/32	7- 3/16	7-23/32	7- 1/16	7-45/64	7- 1/16	7-23/32	6-15/16	7-23/32	6-7/8	7-25/32
76	7-3/8	7-25/32	7- 5/16	7-25/32	7- 5/16	7-27/32	7- 5/16	7-27/32	7- 3/16	7-53/64	7- 3/16	7-27/32	7- 1/16	7-27/32	7	7-29/32
77	7-1/2	7-29/32	7- 7/16	7-29/32	7- 7/16	7-31/32	7- 7/16	7-31/32	7- 5/16	7-61/64	7- 5/16	7-31/32	7- 3/16	7-31/32	7-1/8	8- 1/32
80	7-5/8	8- 1/32	7- 9/16	8- 1/32	7- 9/16	8- 3/32	7- 9/16	8- 3/32	7- 7/16	8- 5/64	7- 7/16	8- 3/32	7- 5/16	8- 3/32	7-1/4	8- 5/32
85									8- 1/16	8-45/64						

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NOTE 1: This drawing was approved by Joint action of the Air Force and Navy Departments on the Five-Navy standard for this product. This drawing supersedes all association standard drawings for the same product and shall become obsolete for the procurement of commercial supplies, or for use in new design, not later than 6 months after the latest date of approval shown.

APPROVED 15 Jan 53 REVISED (3) 4 Apr 55 (5) FOR CHANGES SEE SHEET 1. (6) 4 Aug 82

PROCUREMENT SPECIFICATION	AIR FORCE-NAVY AERONAUTICAL STANDARD	AN42 THRU AN49
MIL-B-6812	BOLT, EYE	SHEET 2 of 2

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